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# A Deep Recurrent Survival Model for Unbiased Ranking

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Slides: [https://jiaruijin.com/project/files/dsr\\_cmb.pdf](https://jiaruijin.com/project/files/dsr_cmb.pdf)

# PART I: Introduction



# Explicit & Implicit Feedback

- **Explicit** Feedback

- score (relevance)
- predict **relevance probability**  $P(r)$  **relevance**

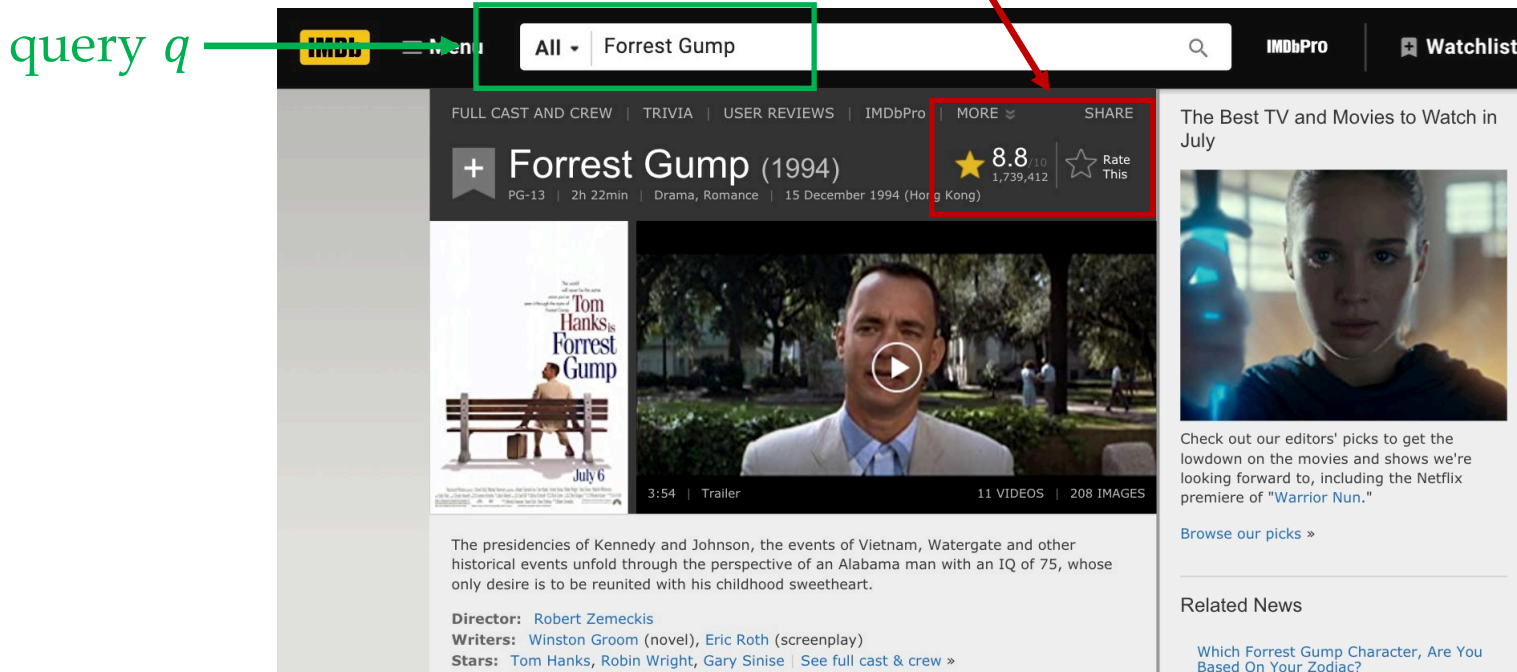


Figure: IMDB Movie



# Explicit & Implicit Feedback

- **Implicit** Feedback
  - click probability, observe probability
  - predict ?

The image shows a Google search interface for the query "university college london". The search bar is highlighted with a green box and labeled "query  $q$ ". Below the search bar, the results are displayed. The first result, "UCL - London's Global University", is highlighted with a red box and labeled "observe  $P(o)$ ". To the left of this result is a red eye icon. Below the first result, there is a "People also ask" section with four questions. The second result, "Undergraduate prospectus 2021 - UCL - University College ...", is also highlighted with a red box and labeled "click  $P(c)$ ". To the left of this result is a red mouse cursor icon. The "People also ask" section is also highlighted with a red box.

Google university college london query  $q$

About 1,390,000,000 results (0.87 seconds)

UCL - London's Global University

UCL is the number one London university for Research Strength (REF2014), recognised for its academic excellence and global impact.

Undergraduate prospectus · UCL Graduate degrees · Prospective students · Staff

People also ask

Is UCL a top university?

What does University College London Specialise in?

Is it hard to get into University College London?

What is UCL acceptance rate?

Feedback

www.ucl.ac.uk › prospective-students › undergraduate

Undergraduate prospectus 2021 - UCL - University College ...

As well as being a top-ranking university, UCL is right in the centre of London so it will enrich you culturally as well as academically. Coronavirus: advice for our ...

Figure: Google





# Position Bias

- Traditional Ranking

- information we have: **click**, **observe**
- predict click and rank click probability  $P(c)$

## Search Tea



Figure: Illustration of position bias



# Position Bias

- Traditional Ranking

- information we have: **click, observe**
- predict click and rank click probability  $P(c)$

- Position Bias

- item at low position is not likely to be observed and, of course, not be clicked

- Unbiased Ranking

- information we have: **click, observe**
- information we need: **relevance probability**  $P(r)$

## Search Tea



Figure: Illustration of position bias



# Unbiased Ranking

- User Browsing Behavior Model
  - provides several basic **assumptions** about user browsing behavior.
  - [Thorsten Joachims, et al.] only model user behavior patterns without **simultaneously sufficient optimization for ranking algorithm**.
- Counterfactual Learning Framework
  - treats the position bias as the **counterfactual factor**.
  - debiases the user feedbacks through **inverse propensity weighting**.
  - ignores the **contextual information** of the given ranking list, e.g., the content of the previous items may influence the observation of the next item.

Ref: Thorsten Joachims, et al. Unbiased learning-to-rank with biased feedback. WSDM, 2017.



contextual information

top  
bottom  
rowse  
rection

click case

non-click  
case



click

unclick

unobserve

observe

untrusted  
observation

observe

click

documents in **future**  
- session are supposed to be  
**unobserved**

when predict

, we involve information of

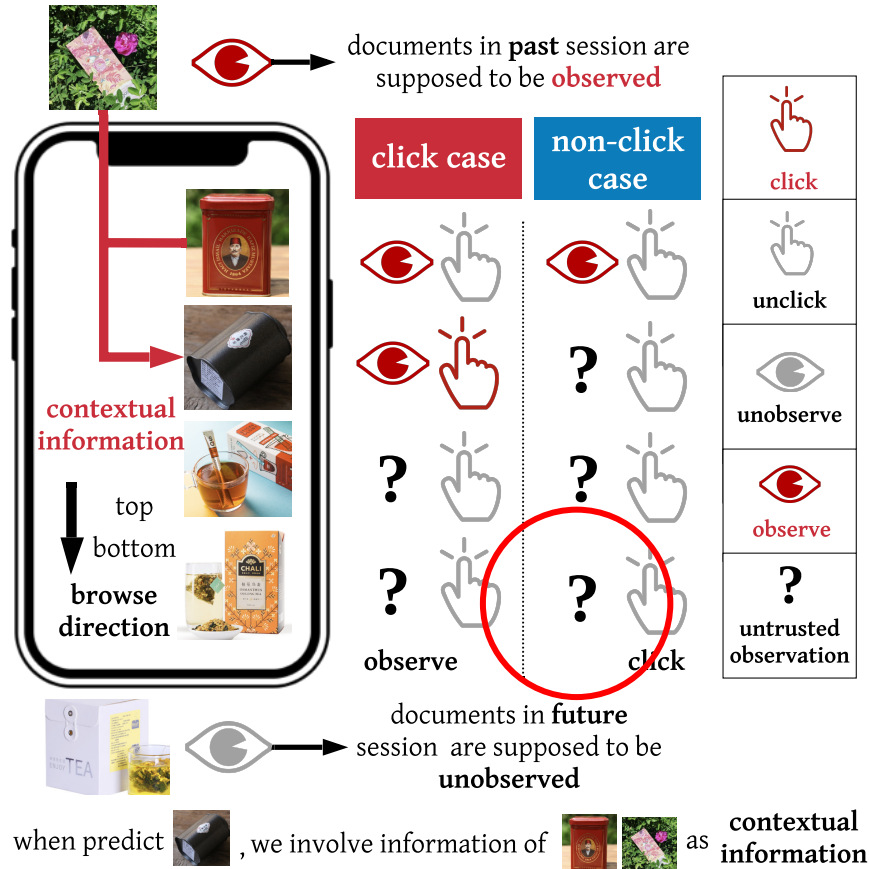
contextual  
information

**Figure:** Illustration of user various behaviors (i.e., click and non-click case) when browsing document list as shown in the left side. Notations are provided in the right side.

[illegible]

**Figure:** Illustration of user various behaviors (i.e., click and non-click case) when browsing document list as shown in the left side. Notations are provided in the right side.

# Current Challenges



(C3) When the user scrolls down the page and observes the presented items, the tracking logs record user observations until the last position. However, this may not be true since the user may stop and lost her attention before that. We called these noisy logs **untrusted observations**.

**Figure:** Illustration of user various behaviors (i.e., click and non-click case) when browsing document list as shown in the left side. Notations are provided in the right side.

# Outline

- Introduction
  - Unbiased Ranking
  - Information Retrieval
- Architecture
  - **Deep Recurrent Survival Model**
  - Point-wise Loss Function
  - Pair-wise Loss Function
- Experiment
- Conclusion

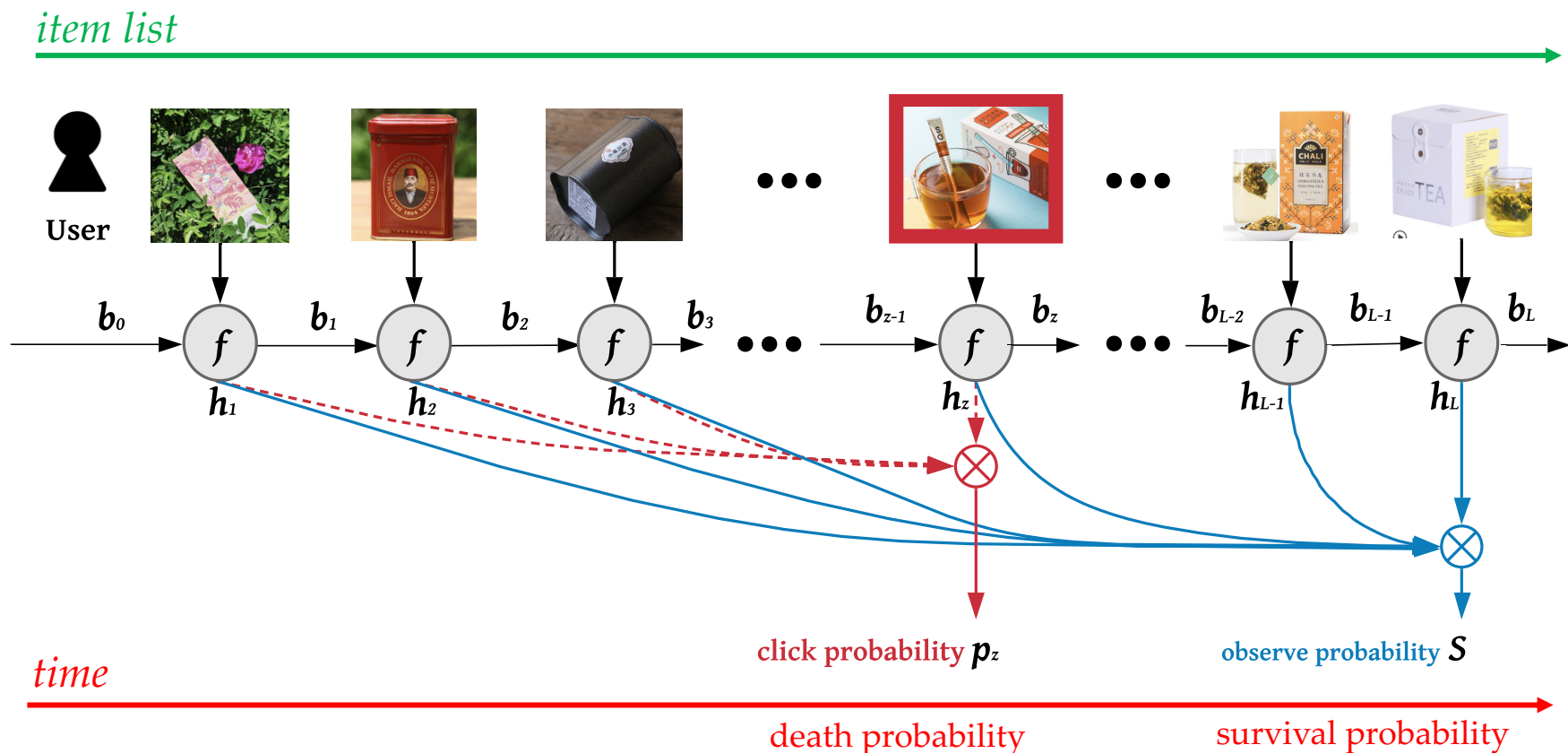


# PART II: Architecture





# Motivation



**Figure:** Illustration of Deep Recurrent Survival Ranking Model. Note that we mine click patterns in click case and observe patterns in both click and non-click cases.

# Survival Model

click event  $z$

- click probability (death probability), the probability density function (P.D.F.) of click occurring at  $i$ -th item

$$p_i = P(c_i = 1) = P(z = i)$$

- observe probability (survival probability), the cumulative distribution function (C.D.F.), since user will keep browsing until she finds and clicks a favored one

$$S(i) = P(o_i = 1) = \sum_{\tau \geq i} P(z = \tau)$$

- relevance probability, the conditional click probability, the click probability at item given that the item is observed

$$h_i = P(r_i = 1) = \frac{P(c_i = 1)}{P(o_i = 1)} = \frac{P(z = i)}{P(z \geq i)} = \frac{p_i}{S_i}$$



# Deep Recurrent Survival Model

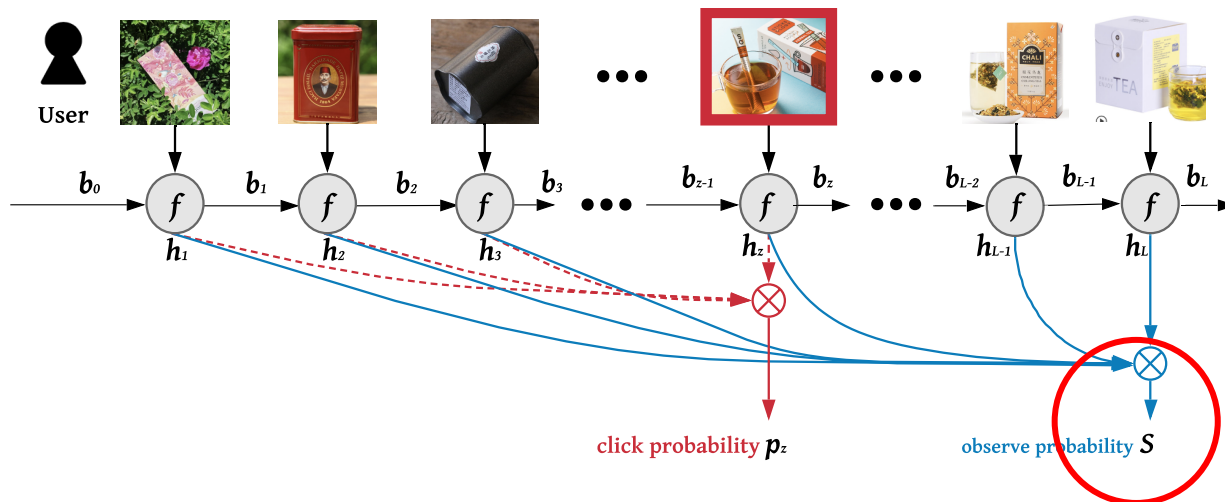
- relevance probability

$$h_i = P(z = i | z \geq i, x; \theta) = f_{\theta}(x_i | b_{i-1})$$

- observe probability (survival probability)

$$\begin{aligned} S(i|x; \theta) &= P(i \leq z) = P(z \neq 1, z \neq 2, z \neq 3, \dots, z \neq i-1 | x; \theta) \\ &= P(z \neq 1 | x_1; \theta) \cdot P(z \neq 2 | z \neq 1, x_2; \theta) \cdots P(z \neq i-1 | z \neq 1, \dots, z \neq i-2 | x_{i-1}; \theta) \end{aligned}$$

$$= \prod_{\tau: \tau < i} [1 - P(z = \tau | z \geq \tau, x_{\tau}; \theta)] = \prod_{\tau: \tau < i} (1 - h_{\tau})$$



# Deep Recurrent Survival Model

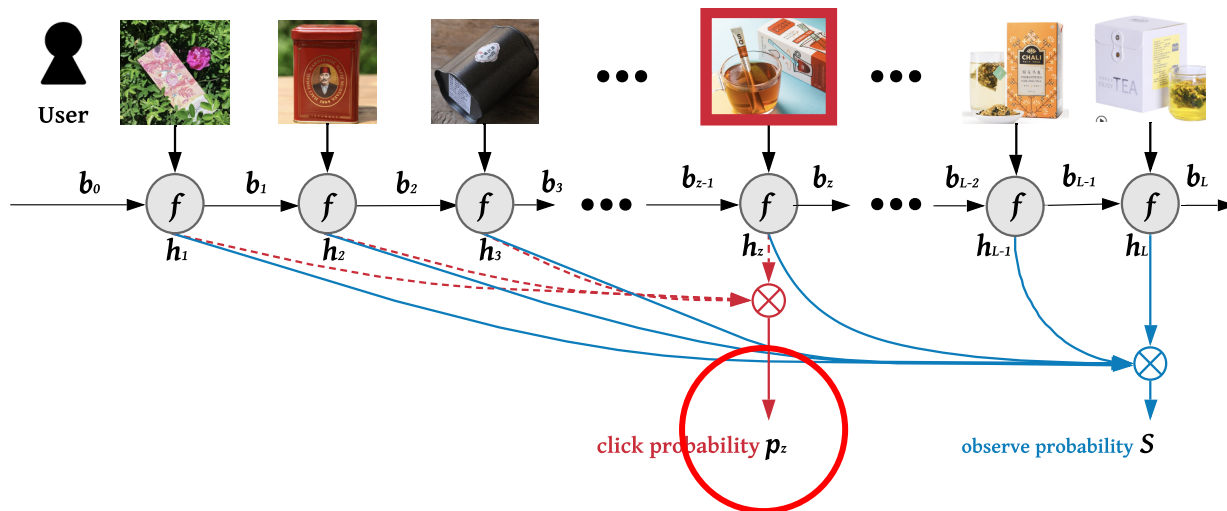
- relevance probability

$$h_i = P(z = i | z \geq i, x; \theta) = f_{\theta}(x_i | b_{i-1})$$

- click probability (death probability)

$$p_i = [1 - S(i + 1)] - [1 - S(i)] = S(i) - S(i + 1)$$

$$p_i = P(z = i | x; \theta) = h_i \prod_{\tau: \tau < i} (1 - h_{\tau})$$



# Point-wise Loss Function

- click probability (P.D.F.)

- aims to **minimize** negative log-likelihood of the click item  $d_j$  over the clicked logs as

$$\begin{aligned} L_{\text{point}(z)} &= -\log \prod_{(x,z) \in \mathcal{D}_{\text{click}}} P(z = j|x; \theta) = -\log \prod_{(x,z) \in \mathcal{D}_{\text{click}}} p_j \\ &= -\log \prod_{(x,z) \in \mathcal{D}_{\text{click}}} \left[ h_j \prod_{\tau: \tau < i} (1 - h_\tau) \right] \\ &= - \sum_{(x,z) \in \mathcal{D}_{\text{click}}} [\log h_j + \sum_{\tau: \tau < i} \log(1 - h_\tau)] \end{aligned}$$



# Point-wise Loss Function

- observe probability (C.D.F.)
  - push up the observe probability of items whose position in range of  $[0, l]$ , and pull down in range of  $[l, \infty]$ .

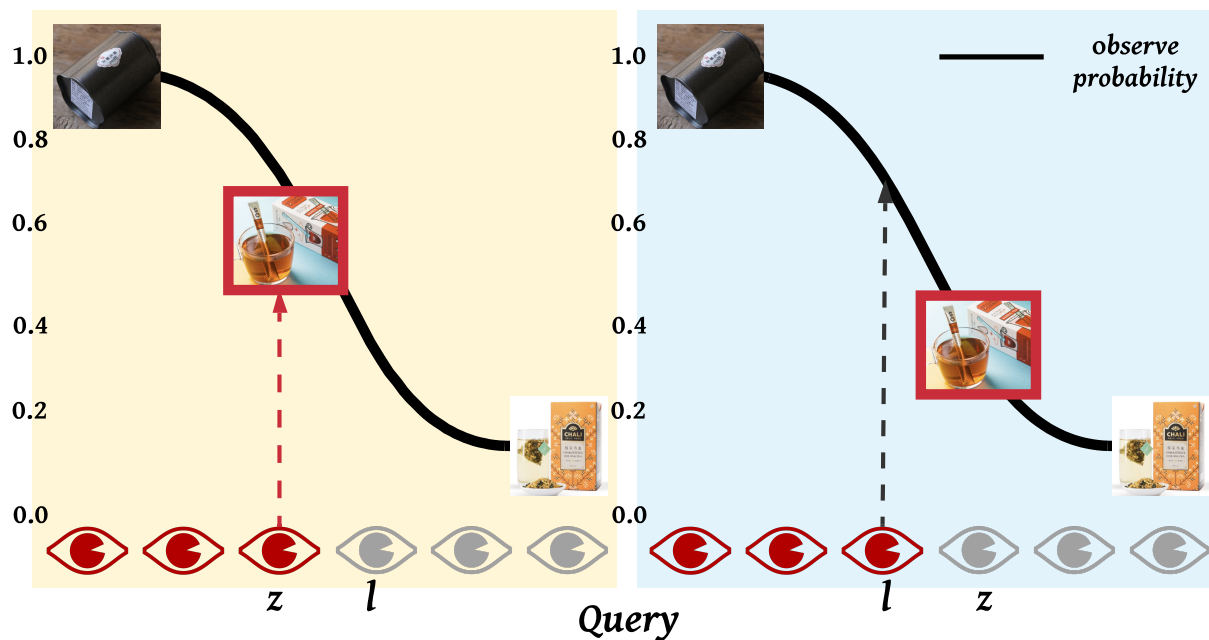


Figure: Intuition behind C.D.F. losses. The left and right sub-figures denote **click** and **non-click** cases respectively.

# Point-wise Loss Function

- observe probability (C.D.F.)

- over the **click cases**

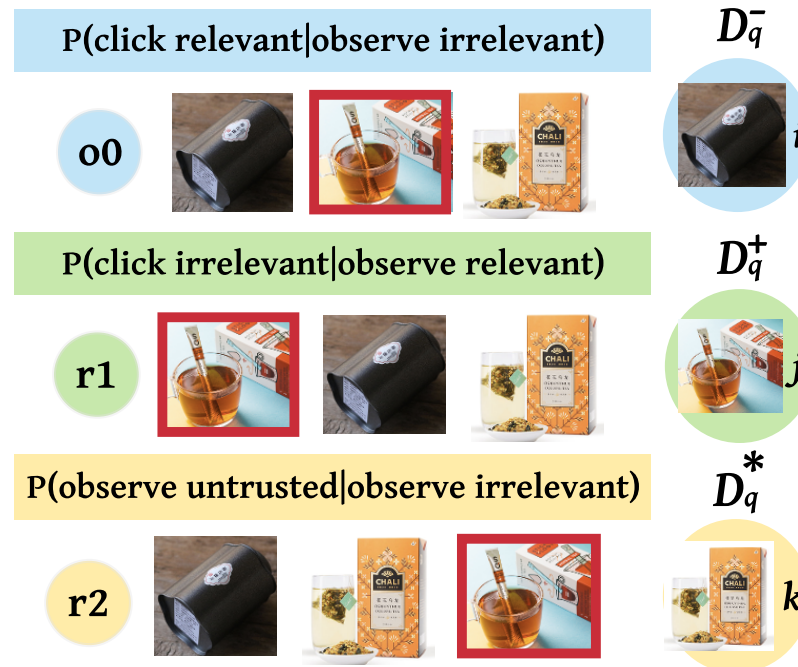
$$\begin{aligned} L_{\text{click}} &= -\log \prod_{(x,l) \in \mathcal{D}_{\text{click}}} P(l \geq z|x; \theta) \approx -\log \sum_{(x,l) \in \mathcal{D}_{\text{click}}} [1 - S(l|x; \theta)] \\ &= - \sum_{(x,l) \in \mathcal{D}_{\text{click}}} \log [1 - \prod_{\tau: \tau < l} (1 - h_{\tau})] \end{aligned}$$

- over the **non-click cases**

$$\begin{aligned} L_{\text{non-click}} &= -\log \prod_{(x,l) \in \mathcal{D}_{\text{non-click}}} P(z > l|x; \theta) \approx -\log \sum_{(x,l) \in \mathcal{D}_{\text{non-click}}} S(l|x; \theta) \\ &= - \sum_{(x,l) \in \mathcal{D}_{\text{non-click}}} \sum_{\tau: \tau < l} \log(1 - h_{\tau}) \end{aligned}$$



# Pair-wise Loss Function



**Figure:** Illustration of permutation item modeling, where items here are sampled from different subsets in the right side.



# Pair-wise Loss Function

- over the **o0 query**

$$\begin{aligned}
 L_{\text{pair}(o0)} &= -\log \prod_{(d_i, d_j) \in I_q} P(z = j | z \geq i, x; \theta) = -\log \prod_{(d_i, d_j) \in I_q} \frac{P(z = j | x; \theta)}{P(z \geq i | x; \theta)} \\
 &= - \sum_{(d_i, d_j) \in I_q} \left\{ \left[ \log h_j \sum_{\tau: \tau < j} \log(1 - h_\tau) \right] - \sum_{\tau: \tau < i} \log(1 - h_\tau) \right\}
 \end{aligned}$$

- over the **r1 query**

$$\begin{aligned}
 L_{\text{pair}(r1)} &= \log \prod_{(d_i, d_j) \in I_q} P(z = i | z \geq j, x; \theta) = \log \prod_{(d_i, d_j) \in I_q} \frac{P(z = i | x; \theta)}{P(z \geq j | x; \theta)} \\
 &= \sum_{(d_i, d_j) \in I_q} \left\{ \left[ \log h_i \sum_{\tau: \tau < i} \log(1 - h_\tau) \right] - \sum_{\tau: \tau < j} \log(1 - h_\tau) \right\}
 \end{aligned}$$



# Pair-wise Loss Function

- over the **r2 query**

$$\begin{aligned} L_{\text{pair}(o0)} &= -\log \prod_{(d_i, d_k) \in I_q} P(z \geq k | z \geq i, x; \theta) = -\log \prod_{(d_i, d_j) \in I_q} \frac{P(z \geq k | x; \theta)}{P(z \geq i | x; \theta)} \\ &= - \sum_{(d_i, d_k) \in I_q} \left\{ \sum_{\tau: \tau < k} \log(1 - h_\tau) - \sum_{\tau: \tau < i} \log(1 - h_\tau) \right\} \end{aligned}$$



# PART III: Experiment



# Experiment

**Table 2: Comparison of different unbiased learning-to-rank methods under Yahoo Search Engine and Alibaba Recommender System. CCM is utilized as click generation model. \* indicates p-value < 0.001 in significance test vs the best baseline.**

| Ranker      | Debiasing Method            | Yahoo Search Engine (CCM) |               |               |               | Alibaba Recommender System (CCM) |               |               |               |
|-------------|-----------------------------|---------------------------|---------------|---------------|---------------|----------------------------------|---------------|---------------|---------------|
|             |                             | MAP                       | NDCG@1        | NDCG@3        | NDCG@5        | MAP                              | NDCG@1        | NDCG@3        | NDCG@5        |
| DRSR (Ours) | Labeled Data                | 0.861                     | 0.747         | 0.759         | 0.771         | 0.850                            | 0.737         | 0.741         | 0.755         |
|             | Pairwise Debiasing          | <b>0.842*</b>             | <b>0.719*</b> | <b>0.721*</b> | <b>0.737*</b> | <b>0.831*</b>                    | <b>0.684*</b> | <b>0.685*</b> | <b>0.707*</b> |
|             | Pointwise Debiasing         | <b>0.839*</b>             | <b>0.713*</b> | <b>0.717*</b> | <b>0.730*</b> | <b>0.830*</b>                    | <b>0.682*</b> | <b>0.684*</b> | <b>0.706*</b> |
|             | Regression-EM [43]          | 0.829                     | 0.679         | 0.685         | 0.701         | 0.820                            | 0.657         | 0.668         | 0.673         |
|             | Click Data                  | 0.817                     | 0.636         | 0.652         | 0.667         | 0.810                            | 0.613         | 0.627         | 0.658         |
| LambdaMART  | Labeled Data                | 0.854                     | 0.745         | 0.745         | 0.757         | 0.847                            | 0.729         | 0.732         | 0.743         |
|             | Ratio Debiasing [17]        | 0.830                     | 0.688         | 0.685         | 0.699         | 0.821                            | 0.661         | 0.669         | 0.674         |
|             | Regression-EM [43]          | 0.826                     | 0.669         | 0.676         | 0.691         | 0.818                            | 0.636         | 0.651         | 0.667         |
|             | Click Data                  | 0.813                     | 0.628         | 0.646         | 0.673         | 0.804                            | 0.603         | 0.618         | 0.646         |
| DNN         | Labeled Data                | 0.831                     | 0.677         | 0.685         | 0.705         | 0.824                            | 0.674         | 0.679         | 0.693         |
|             | Dual Learning Algorithm [2] | 0.825                     | 0.672         | 0.678         | 0.691         | 0.814                            | 0.629         | 0.647         | 0.674         |
|             | Regression-EM [43]          | 0.823                     | 0.665         | 0.669         | 0.687         | 0.813                            | 0.628         | 0.645         | 0.672         |
|             | Click Data                  | 0.809                     | 0.611         | 0.619         | 0.648         | 0.801                            | 0.600         | 0.612         | 0.641         |

[Regression-EM] [Xuanhui Wang, et al.] a regression-based EM method where position bias is estimated from regular clicks.

[Dual Learning Algorithm] [Qingyao Ai, et al.] a dual learning framework jointly learning a ranker and debiasing click data.

[Ratio Debiasing] [Ziniu Hu, et al.] an unbiased pair-wise learning-to-rank based on inverse propensity weight.

[Point-wise Debiasing] our model in point-wise setting.

[Pair-wise Debiasing] our model in pair-wise setting.

[Click Data] the raw click data without debiasing to train the ranker.

[Labeled Data] the human annotated relevance labels without any bias.

Ref: Xuanhui Wang, et al. Position bias estimation for unbiased learning to rank in personal search. WSDM, 2018.

Ref: Qingyao Ai, et al. Unbiased Learning to Rank with Unbiased Propensity Estimation. SIGIR, 2018.

Ref: Ziniu Hu, et al. Unbiased LambdaMART: An Unbiased Pairwise Learning-to-Rank Algorithm. WWW, 2019.



# Experiment

Table 3: Comparison with PBM as click generation model.  
Notations are same with Table 2.

| Yahoo Search Engine (PBM) |               |               |               |               |
|---------------------------|---------------|---------------|---------------|---------------|
| Ranker                    | MAP           | NDCG@1        | NDCG@3        | NDCG@5        |
| DRSR (Ours)               | 0.861         | 0.747         | 0.759         | 0.771         |
|                           | <b>0.848*</b> | <b>0.726*</b> | <b>0.737*</b> | <b>0.745*</b> |
|                           | <b>0.843*</b> | <b>0.723*</b> | <b>0.731*</b> | <b>0.740*</b> |
|                           | 0.834         | 0.698         | 0.705         | 0.712         |
|                           | 0.825         | 0.671         | 0.679         | 0.693         |
| LambdaMART                | 0.854         | 0.745         | 0.745         | 0.757         |
|                           | 0.836         | 0.717         | 0.716         | 0.728         |
|                           | 0.830         | 0.685         | 0.684         | 0.700         |
|                           | 0.820         | 0.658         | 0.669         | 0.672         |
| DNN                       | 0.831         | 0.677         | 0.685         | 0.705         |
|                           | 0.828         | 0.674         | 0.683         | 0.697         |
|                           | 0.829         | 0.676         | 0.684         | 0.699         |
|                           | 0.819         | 0.637         | 0.651         | 0.667         |

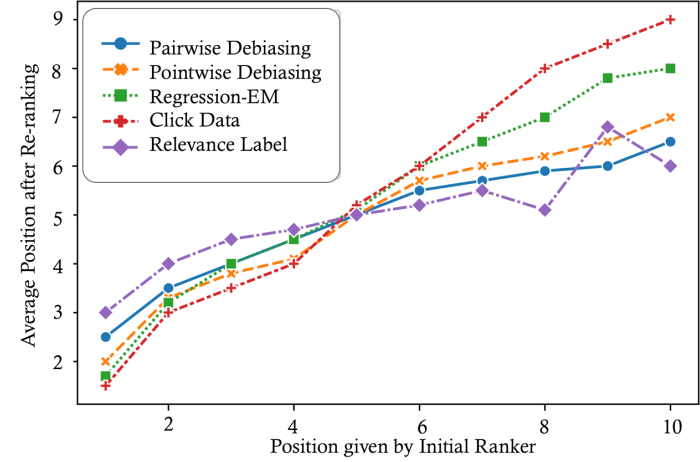


Figure: Average positions after re-ranking of documents at each original position by different debiasing methods combined with DRSR.

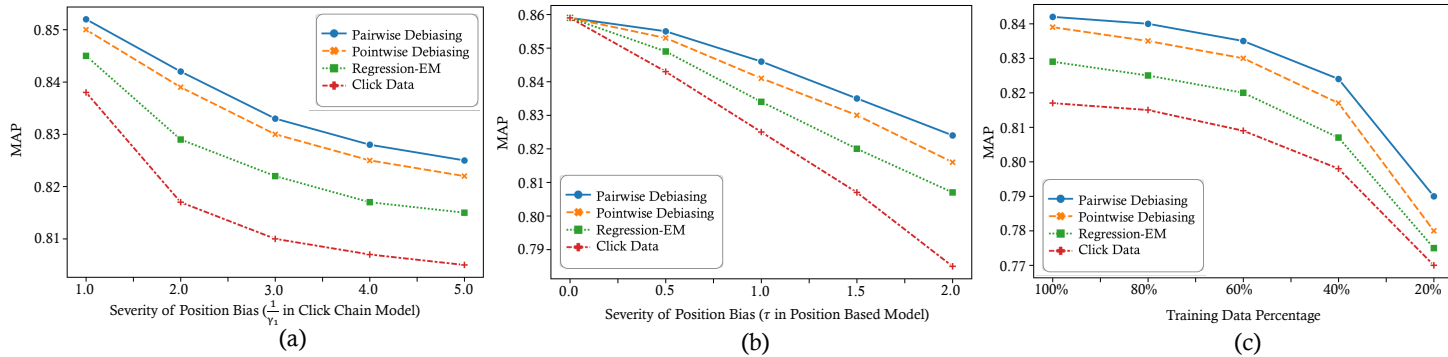


Figure: (a) & (b) Performance of DRSR against other debiasing methods with different degrees of position bias. (c) with different sizes of training data.



# PART IV: Conclusion



# Conclusion

- We adopt **survival analysis techniques** accompanied with **probability chain rule** to derive the joint probability of user various behaviors.
- This framework enables unbiased model to leverage the **contextual information** in the ranking list.
- Also, we incorporate with survival analysis, and thus can model the **non-click queries** as the **censored click logs**.
- DRSR can be easily adopted in the **pair-wise** loss setting to capture relative relevance between trusted function and **untrusted observation**.

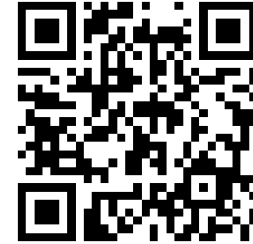


# Thanks for Your Listening



Slides Link

Jerry's Wechat



Full Paper Link

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Website: <https://jjaruijin.com/>